

# PULKIT BANSAL

· bansalpulk0812@gmail.com · [LinkedIn](#) · [GitHub](#) · EN / FR Bilingual · PGWP Holder

## TECHNICAL PROFILE

Software engineer (M.Eng. Software Engineering, Concordia University, 2026) with full-stack and systems experience across Java, Spring Boot, Angular, and Python. Builds scalable REST APIs, distributed backends, and CI/CD pipelines in Agile teams. Work-authorized in Canada (PGWP). Bilingual EN/FR, fast learner, strong communicator.

## TECHNICAL SKILLS

**Frameworks:** Spring Boot · Angular · Hibernate · Flask · FastAPI · Flutter · Node.js

**Backend & APIs:** Spring Boot · Flask · FastAPI · RESTful · microservices · OOP design patterns

**Languages:** Java · Python · C++ · TypeScript · JavaScript · SQL  
**Databases & Pipelines:** PostgreSQL · Oracle · Hibernate · ETL · time-series schemas

**DevOps & Tooling:** Docker · GitHub Actions · CI/CD · Git · Linux CLI

## PROJECTS

**QuantFrame — Algo Trading Data Pipeline & Backtesting Engine** · Personal / Academic

2025 – Present

*Python · asyncio · C++ · PostgreSQL · Docker · Linux*

- Async market data ingestion from Alpaca REST API (10+ symbols, OHLCV + NBBO quotes) with concurrent batch fetching and rate-limit backoff.
- Partitioned PostgreSQL schema (year-range partitions, BRIN indexes, composite PK) — sub-10 ms range-scan on 2-year datasets.
- Incremental gap-fill orchestrator: reads MAX(ts) per symbol before each run — idempotent, safe to restart after failure.
- C++ lock-free order book engine (in progress): nanosecond timestamps, pluggable OOP strategy classes for backtesting.
- CLI (ingest | status | bars), Docker Compose stack, full pytest suite with mocked HTTP; CI on Linux via GitHub Actions.

**ClaimsAI — AI-Powered Insurance Claims Management Platform** · Personal Project

2025 – 2026

*Java · Spring Boot · Angular · Spring AI · Kafka · Elasticsearch · PostgreSQL · Docker*

- Architected 6 Spring Boot microservices (Auth, Policy, Claims, AI, Search, API Gateway) with event-driven communication via Apache Kafka and JWT/OAuth2 security.
- Built RAG pipeline using Spring AI + pgvector for AI-powered claims chat, fraud detection (risk scoring 0.0–1.0), and automated claim summarization.
- Implemented Coveo-style semantic search with Elasticsearch multi-match queries, fuzzy matching, and MongoDB audit trail for compliance.
- Containerized full stack (14 containers) with Docker Compose; one-command deployment with health-check dependencies and multi-stage builds.

## WORK EXPERIENCE

**Full Stack Developer Intern** · Tech Mahindra COE, India

Jan 2024 – Jun 2024

- Shipped RESTful APIs (Spring Boot/Python microservices) improving throughput by 20%; applied OOP design patterns (strategy, repository, factory) across large Java codebase.
- Automated CI/CD with GitHub Actions + Docker, cutting deployment time by 40%; led code reviews resolving 30+ critical issues in Agile sprints.

**Flutter Android Developer Intern** · Uddharana Tech Pvt. Ltd., India

Aug 2022 – Oct 2022

- Built and shipped a Flutter + Firebase Android app to Google Play Store; designed Firestore data models and UX for farmer/merchant profiles.

## EDUCATION

**M.Eng. Software Engineering CO-OP**

Concordia University, Montréal, QC • Sept 2024 – June 2026

Core Exec, Graduate Students Association (2025–26)

**B.Sc. Computer Science and Engineering**

Chitkara University, India • Oct 2020 – Aug 2024

VP, ACM Student Chapter (2022–23) · GPA: honors